

## **Subject Description Form**

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| <b>Subject Code</b>                                          | <b>ITC3206T</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Subject Title</b>                                         | <b>Apparel Production Engineering</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Credit Value</b>                                          | <b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Level</b>                                                 | <b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pre-requisite / Co-requisite/ Exclusion</b>               | ITC2204T Foundations of Apparel Construction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Objectives</b>                                            | The subject provides technical knowledge of apparel production processes and their respective technical problems during manufacturing. It evaluates and enhances the efficiency of production processes for cost-effectiveness. It also identifies the problems encountered in apparel manufacturing and investigates their solutions in response to the changing demands of the industry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Intended Learning Outcomes</b><br><br>(Note 1)            | Upon completion of the subject, students will be able to: <ul style="list-style-type: none"> <li>a. Manage, apply and evaluate the production techniques and processes in apparel manufacturing, including spreading, cutting, sewing, etc. in relation to material utilisation, production efficiency, quality and costing of apparel products.</li> <li>b. Demonstrate an integrated and critical approach to the processes and technologies involved in apparel production so as to solve problems and achieve the required quality standards within an ever changing international and local market environment.</li> <li>c. Proactively and critically evaluate the new technologies and production methods from an industrial engineer perspective and within the context of global developments and trends.</li> </ul>                                                                                                                                                                            |
| <b>Subject Synopsis/ Indicative Syllabus</b><br><br>(Note 2) | <p><b>(I) Cutting</b></p> <ul style="list-style-type: none"> <li>-Principal processes in the cutting room, cut order planning, spreading, cutting, sorting, material utilization in relation to apparel production</li> <li>-Evaluation of material usage / wastage calculation</li> <li>-Application of cutting technology and integration of computer systems in the cutting room</li> <li>-Analysis of the quality, costs and productivity of cutting room processes</li> </ul> <p><b>(II) Sewing</b></p> <ul style="list-style-type: none"> <li>-Sequence of assembly and flow of operations for various types of apparel products</li> <li>-Appreciation of the characteristics of various apparel production systems</li> <li>-Application of specific sewing machinery and attachments for apparel production</li> <li>-Quality aspects and specification of sewing operations</li> <li>-Selection of making-up methods and seams for apparel products of various quality and end-uses</li> </ul> |

|                                                                                            | <p><b>(III) Industrial Engineering</b></p> <ul style="list-style-type: none"><li>-Major production concepts of work flow, balancing, throughput, cycle time, work-in-process</li><li>-Line balancing techniques used in production: strengths and limitations</li><li>-Principal and application of Standard Minute Value (SMV) in cost and production planning</li><li>-Optimization of production scheduling and capacity planning</li><li>- Fundamentals of method study, workplace engineering and time measurement</li><li>-Techniques and systems for elimination of material handling</li><li>-Applications of engineering, psychological, and physiological principles to the analysis and ergonomic design of human work systems</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                   |             |                                                                                |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |
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| <p><b>Teaching/Learning Methodology</b><br/>(Note 3)</p>                                   | <p>Lectures are to provide the general framework and principles of various apparel production processes. Professional seminars will help students to build their first-hand knowledge and related technical problems in specific production processes. Tutorials will be used to supplement formal lectures and seminars for further exploration of concepts and principles. Videos and/or visits will be employed to enhance students’ understanding of the latest technologies and industrial practice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |             |                                                                                |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |
| <p><b>Assessment Methods in Alignment with Intended Learning Outcomes</b><br/>(Note 4)</p> | <table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th></th><th></th><th></th></tr><tr><td>1. Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> <p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</p> <p>The assessment instruments will include a group project that demonstrates students’ abilities to exploit and integrate the processes and technologies, tackle quality problems, as well as evaluate new developments in the market. The knowledge building worksheet and in-class discussion during tutorials aim to provide students a better understanding of work-based environment and real-life problem scenarios in the production processes. Students’ performance in terms of problem-solving techniques, critical thinking and analytical skills to be applied in the industry will also be assessed by means of a written examination.</p> <p><i>The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of</i></p> | Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |  |  |  |  |  | a | b | c |  |  |  | 1. Continuous Assessment | 50% | ✓ | ✓ | ✓ |  |  |  | 2. Examination | 50% | ✓ | ✓ | ✓ |  |  |  | Total | 100 % |  |  |  |  |  |  |
| Specific assessment methods/tasks                                                          | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                   |             | Intended subject learning outcomes to be assessed (Please tick as appropriate) |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |
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| 1. Continuous Assessment                                                                   | 50%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ✓                                 | ✓           | ✓                                                                              |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |
| 2. Examination                                                                             | 50%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ✓                                 | ✓           | ✓                                                                              |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |
| Total                                                                                      | 100 %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                   |             |                                                                                |  |  |  |  |  |   |   |   |  |  |  |                          |     |   |   |   |  |  |  |                |     |   |   |   |  |  |  |       |       |  |  |  |  |  |  |

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|                                      | <b><i>academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Student Study Effort Expected</b> | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|                                      | ▪ Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 26 Hrs.  |
|                                      | ▪ Studio/Tutorial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12 Hrs.  |
|                                      | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
|                                      | ▪ Projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 66 Hrs.  |
|                                      | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 104 Hrs. |
| <b>Reading List and References</b>   | <p><b><u>Books</u></b><br/> Carr, H, and Latham, B. (2000), <i>The Technology of Clothing Manufacture</i>. Blackwell Scientific Publications, Oxford.</p> <p>Chuter, A.J. (1995), <i>Introduction to Clothing Production Management</i>. BSP Professional Books.</p> <p>Chuter, A.J. (2002), <i>Quality Management in the Clothing and Textile Industries</i>. Textile Institute.</p> <p>Crowson R. (2006), <i>Factory Operations: Planning and Instructional Methods</i>. CRC Press,US.</p> <p>Fairhurst C. (2008), <i>Advances in Apparel Production</i>. Woodhead, Cambridge.</p> <p>Hunter, A., King, R. and Lowson, R.H. (2002), <i>The Textile/Clothing Pipeline and Quick Response Management</i>. Textile Institute, c2002, Manchester, England.</p> <p>Meyers, F.E. and Stewart, J.R. (2002), <i>Motion and Time Study for Lean Manufacturing</i>. 3<sup>rd</sup> edition, Prentice Hall, Upper Saddle River, N.J.</p> <p>Ramesh, B.V. (2012), <i>Industrial Engineering in Apparel Production</i>. Woodhead, Cambridge.</p> <p>Tyler, D.J. (1991), <i>Materials Management in Clothing Production</i>. BSP Professional Books.</p> <p><b><u>Supplementary</u></b><br/> Glock, Ruth E. (2005), <i>Apparel Manufacturing: Sewn Product Analysis</i>. 4<sup>th</sup> edition, Prentice Hall, Upper Saddle River, N.J.</p> <p>Lausch, G. &amp; Chazin, M. (2003), “Move up to Improved production”, <i>Upholstery Design and Management</i>, 16(10): 54-56.</p> <p>Salvendy, G. (2001), <i>Handbook of Industrial Engineering: Technology</i></p> |          |

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|  | <p><i>and Operations Management</i>. 3<sup>rd</sup> edition, Wiley, New York, NY.</p> <p><b><u>Websites:</u></b></p> <p>Sonobondultrasonics<br/> <a href="http://www.sonobondultrasonics.com">http://www.sonobondultrasonics.com</a></p> <p>Gerber technology<br/> <a href="http://www.gerbertechnology.com">http://www.gerbertechnology.com</a></p> <p>Lectra<br/> <a href="http://www.lectra.com">http://www.lectra.com</a></p> <p>Eastmancuts<br/> <a href="http://www.eastmancuts.com">http://www.eastmancuts.com</a></p> <p>Shimaseiki<br/> <a href="http://www.shimaseiki.co.jp">http://www.shimaseiki.co.jp</a></p> |
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**Note 1: Intended Learning Outcomes**

Intended learning outcomes should state what students should be able to do or attain upon completion of the subject. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

**Note 2: Subject Synopsis/ Indicative Syllabus**

The syllabus should adequately address the intended learning outcomes. At the same time over-crowding of the syllabus should be avoided.

**Note 3: Teaching/Learning Methodology**

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

**Note 4: Assessment Method**

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method purports to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing